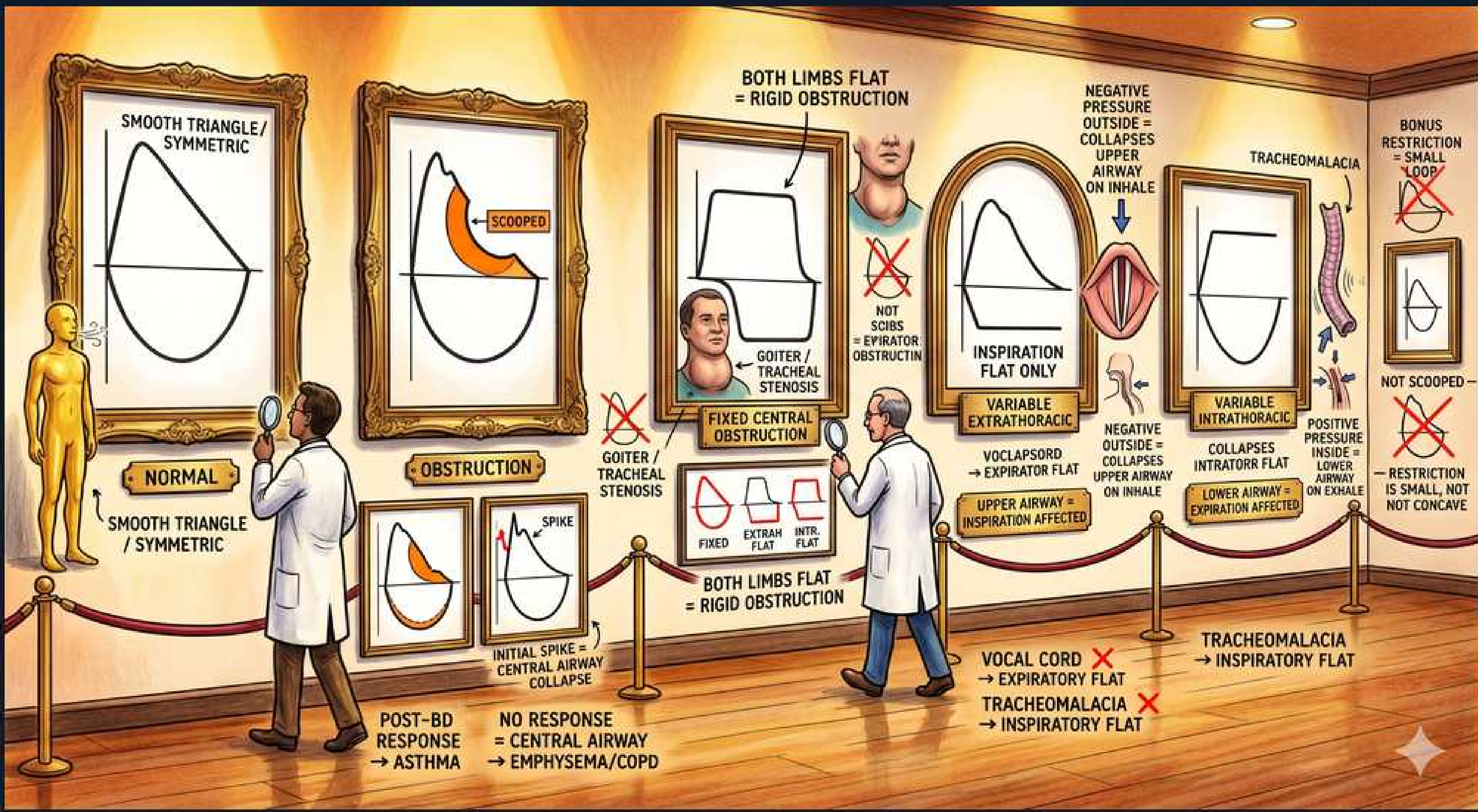
Respiratory Disturbance — Flow-Volume Loop Gallery (Upper-Airway Obstruction)



1. Normal loop

WHAT YOU SEE

First framed tracing: smooth symmetric triangle

WHAT IT ENCODES

The normal flow-volume loop has a rapid rise to a sharp PEAK expiratory flow followed by a straight, gently downsloping expiratory limb, and a rounded (semicircular) inspiratory limb below. Expiration is plotted on top, inspiration on the bottom; the x-axis is lung volume (TLC at left, RV at right).

BOARD TRAP

Loss of the symmetric peaked shape — flattening (plateau) or scooping — is the first clue to obstruction or upper-airway disease; read the SHAPE before any numbers.

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2. Normal label

WHAT YOU SEE

'NORMAL — smooth triangle / symmetric' caption

WHAT IT ENCODES

Use the normal loop as the reference standard: peaked expiratory limb, rounded inspiratory limb, both unflattened. The inspiratory limb is effort-dependent and normally symmetric to expiration in the mid-portion.

BOARD TRAP

A normal FEV1/FVC ratio can still hide a flattened limb of upper-airway obstruction — central/upper-airway lesions distort the loop shape without necessarily dropping the ratio, so inspect the contour, not just spirometric numbers.

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3. Obstruction (scooped)

WHAT YOU SEE

Second tracing with concave/scooped expiratory limb

WHAT IT ENCODES

Lower-airway (intrathoracic small-airway) obstruction — asthma and COPD — produces a CONCAVE, 'scooped-out' expiratory limb because flow falls disproportionately at low lung volumes as small airways collapse. RV/TLC rises with air trapping.

BOARD TRAP

Scooping = diffuse SMALL-airway obstruction, NOT central/upper-airway obstruction (which gives flat plateaus). Don't confuse a concave limb with the boxy plateau of a fixed central lesion.

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4. Post-bronchodilator response

WHAT YOU SEE

Caption: 'response → asthma, no response → emphysema/COPD'

WHAT IT ENCODES

Bronchodilator reversibility testing: a rise in FEV1 of ≥12% AND ≥200 mL after a SABA defines significant reversibility, supporting asthma. COPD/emphysema typically shows little or no reversibility (fixed obstruction).

BOARD TRAP

Lack of bronchodilator reversibility points to COPD, not asthma — but a single non-reversible test doesn't exclude asthma entirely (asthma can become partly fixed with remodeling).

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5. Initial spike (central collapse)

WHAT YOU SEE

Small tracing with sharp initial spike then collapse

WHAT IT ENCODES

Emphysema's loop shows a brief, sharp INITIAL expiratory spike followed by rapid airway collapse and a steeply concave limb — the loss of elastic recoil lets airways collapse early in expiration once the initial burst passes.

BOARD TRAP

The early spike-then-collapse is emphysema's SMALL-airway pattern; it is distinct from the flat-topped PLATEAU of central/upper-airway obstruction.

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6. Fixed central obstruction (both flat)

WHAT YOU SEE

Tracing with BOTH inspiratory and expiratory limbs flattened (box shape)

WHAT IT ENCODES

A FIXED upper-airway/central obstruction (tracheal stenosis, large goiter, fixed mass) flattens BOTH the inspiratory and expiratory limbs, giving a rectangular 'boxed' loop, because the rigid narrowing limits flow equally in both phases regardless of pressure.

BOARD TRAP

Both limbs flat = FIXED lesion. If only ONE limb flattens, it is a VARIABLE obstruction (the limb tells you intra- vs extrathoracic).

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7. Goiter / tracheal stenosis

WHAT YOU SEE

Neck figure labeled goiter / tracheal stenosis

WHAT IT ENCODES

Tracheal stenosis (post-intubation, granulomatosis with polyangiitis) and a large goiter are the classic FIXED central lesions producing the boxed-off loop with flattening of both limbs.

BOARD TRAP

Don't call a fixed central lesion 'asthma' — fixed obstruction flattens both limbs (no scooping) and does not reverse with bronchodilators.

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8. Variable extrathoracic (insp flat)

WHAT YOU SEE

Tracing with flattened INSPIRATORY limb only

WHAT IT ENCODES

Variable EXTRAthoracic obstruction (vocal cord dysfunction, extrathoracic tracheomalacia, extrathoracic tumor) flattens the INSPIRATORY limb only. During inspiration, the airway pressure inside the extrathoracic airway falls below atmospheric, so the narrowing collapses inward.

BOARD TRAP

Extrathoracic = INspiratory flattening. The negative intraluminal pressure of inhalation collapses the extrathoracic airway; on exhalation positive pressure splints it open.

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9. Extrathoracic collapse mechanism

WHAT YOU SEE

Neck diagram: negative pressure outside collapses airway on inhale

WHAT IT ENCODES

Mechanism: the extrathoracic airway sits in atmospheric pressure. On inspiration, intraluminal pressure drops below atmospheric → transmural pressure collapses the variable lesion → inspiratory flow plateau. On expiration the airway is pushed open, so expiration is preserved.

BOARD TRAP

Mixing up the mechanism leads to assigning the wrong limb — remember extrathoracic lesions impair INSPIRATION (the limb that flattens).

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10. Vocal cord dysfunction

WHAT YOU SEE

Caption: 'vocal cord → flat' / inspiratory affected

WHAT IT ENCODES

Vocal cord dysfunction (paradoxical vocal fold motion) is a variable EXTRAthoracic lesion → flattened INSPIRATORY limb. It causes episodic stridor/dyspnea that MIMICS asthma but localizes to the throat, with inspiratory adduction of the cords on laryngoscopy.

BOARD TRAP

VCD mimics refractory asthma but does NOT respond to bronchodilators or steroids — the flattened INSPIRATORY limb (and normal lower-airway flows) gives it away. Treat with speech therapy, not escalating asthma meds.

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11. Variable intrathoracic (exp flat)

WHAT YOU SEE

Tracing with flattened EXPIRATORY limb only

WHAT IT ENCODES

Variable INTRAthoracic obstruction (tracheomalacia, intrathoracic tumor/mass, intrathoracic tracheal lesion) flattens the EXPIRATORY limb only. During forced expiration the rising pleural/intrathoracic pressure exceeds intraluminal pressure and collapses the lesion.

BOARD TRAP

Intrathoracic = EXpiratory flattening. Positive pleural pressure during forced exhalation collapses the intrathoracic airway; inspiration (negative pleural pressure) pulls it open.

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12. Tracheomalacia

WHAT YOU SEE

Floppy trachea image labeled TRACHEOMALACIA

WHAT IT ENCODES

Tracheomalacia = a floppy, weakened trachea. When intrathoracic, it is a variable intrathoracic obstruction that collapses on forced EXPIRATION → expiratory limb plateau (and expiratory wheeze/barking cough).

BOARD TRAP

Tracheomalacia flattens EXpiration; do not confuse it with extrathoracic lesions (VCD), which flatten INspiration.

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13. Intrathoracic collapse mechanism

WHAT YOU SEE

Diagram: positive pressure outside collapses lower airway on exhale

WHAT IT ENCODES

Mechanism: intrathoracic airways are surrounded by pleural pressure. During forced expiration pleural pressure becomes strongly positive and exceeds intraluminal pressure at the variable lesion → collapse → expiratory plateau. During inspiration negative pleural pressure holds it open.

BOARD TRAP

Reversing the inspiratory/expiratory rule between intra- and extrathoracic lesions is THE classic board trap: Intrathoracic = Expiration, Extrathoracic = Inspiration, Fixed = both.

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14. Restriction (small, not scooped)

WHAT YOU SEE

Bonus box: small narrow loop, 'restriction — small, not concave'

WHAT IT ENCODES

Restriction shrinks the WHOLE loop into a tall, narrow 'witch's hat' shifted toward low volumes (low TLC and RV), but the expiratory limb keeps a normal convex/peaked shape with NO scooping — flows may even be supranormal for the small volume.

BOARD TRAP

A small loop without a concave limb is RESTRICTION, not obstruction — shape (concavity), not just size, decides. Obstruction scoops; restriction shrinks.
